

Day 4 - Facilitation Guide

(Working with HTML5)

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(2 hrs) ILT

I. Recap of the previous session

In the last session we explained to you with examples how to create a div and table in HTML.

The Div tag is used in HTML to make divisions of content in an HTML page like (text, images, header, footer, navigation bar, etc). Div tag has both open (<div>) and closing (</div>) tag. The <div> tag is used as a container for HTML elements - which can be styled with CSS or manipulated with JavaScript.

We also explained how to create an HTML table. We learned about the utility of table related tags like <TR>, <TD>, <TH> etc. We also discussed how to create an HTML using various form elements.

We also described the following HTML form elements in our last sessions:

- <form>
- <input>
- <button>
- <select>
- <option>

We also learned how to add <label> tags inside the form for improving accessibility and user experience. In this session we will get introduced to HTML5. We will also discuss basics of HTML5 forms along with examples.

II. Introduction to HTML5

HTML5, the fifth version of the Hypertext Markup Language, is the standard language used to structure and present content on the World Wide Web. It was published by the Web Hypertext Application Technology Working Group (WHATWG) and the World Wide Web Consortium (W3C) in October 2014. HTML5 introduces several new features, APIs, and behaviors that enhance the capabilities of web development. In the next section we will explain HTML5 in detail.

III. Overview of HTML5

HTML5 is the latest standard of Hypertext Markup Language, it supports developing web pages using well structured code and helps create an attractive presentation of web pages. It's made up of three parts: HTML, which keeps the structure; CSS, which affects the presentation; and JavaScript, which creates interactive effects within web browsers.

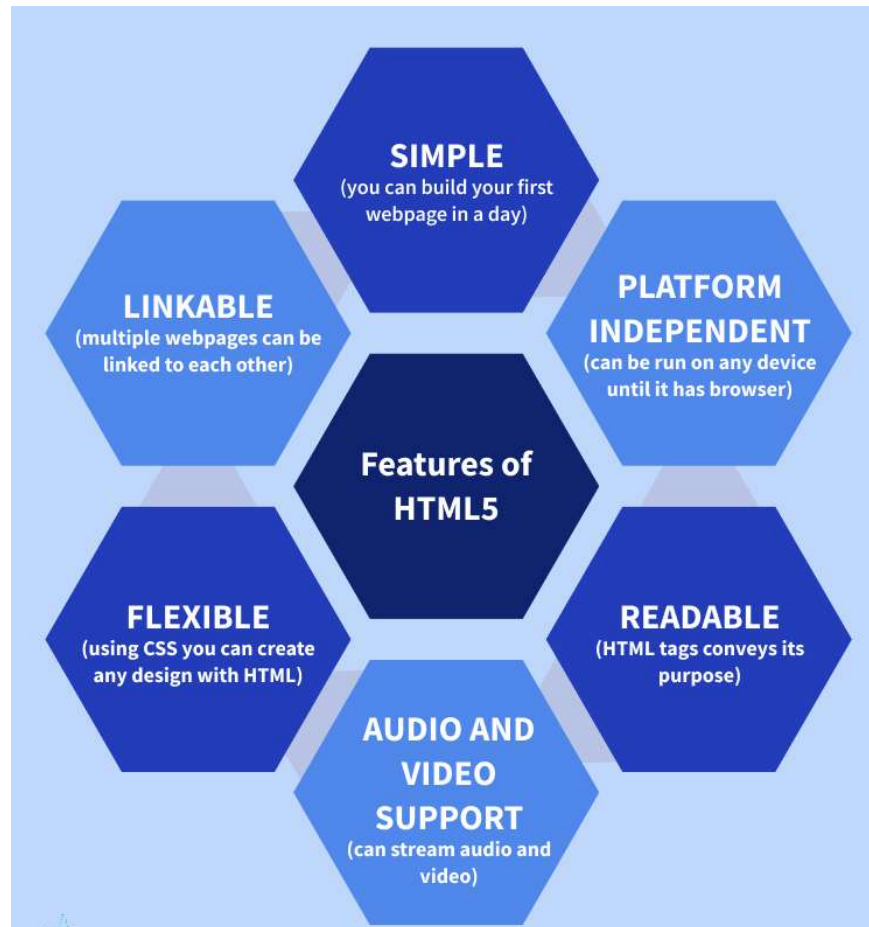
HTML itself is a set of rules that tell web browsers how to interpret code to display web pages. HTML5, the 5th version of this set, improves and upgrades Website capabilities, and helps developers make web content, such as Websites and web applications, more interactive and dynamic.

In this diagram we have listed the important benefits of HTML5.



IV. New features and enhancements in HTML5

HTML5 has introduced many new features over previous versions of HTML. Following diagram represents the overall view of HTML5 features.



Now let us go through various Key Features and Improvements incorporated in HTML5:

Simplified Doctype: HTML5 introduces a simplified and shorter DOCTYPE declaration (`<!DOCTYPE html>`) compared to previous versions.

New Semantic Elements: HTML5 introduces new semantic elements like `<article>`, `<section>`, `<header>`, `<footer>`, `<nav>`, and `<aside>`. These elements provide a clearer structure to web documents, making it easier to understand the content's meaning.

Improved Forms: HTML5 includes new input types (email, url, date, etc.) and attributes (required, placeholder, pattern, etc.) that enhance the functionality and validation of web forms.

Native Audio and Video Support: HTML5 provides `<audio>` and `<video>` elements, allowing seamless embedding of multimedia content without relying on third-party plugins like Flash.

Canvas and SVG Graphics: HTML5 includes the <canvas> element for dynamic, scriptable rendering of 2D shapes and bitmap images. Additionally, SVG (Scalable Vector Graphics) is natively supported for creating vector graphics on the web.

Offline Web Applications: HTML5 introduces technologies like Service Workers and the Application Cache, enabling web applications to work offline and improve performance through caching strategies.

Web Storage: HTML5 provides localStorage and sessionStorage APIs, allowing web applications to store data persistently or temporarily within the user's browser.

Improved Accessibility: HTML5 includes native support for ARIA (Accessible Rich Internet Applications) landmarks, improving web accessibility for users with disabilities.

GeoLocation API: HTML5 includes the Geolocation API, enabling web applications to access the user's geographical location, facilitating location-based services and applications.

Web Workers: HTML5 allows the use of Web Workers, which are background scripts that run parallel to the main page, enabling multi-threaded JavaScript execution for improved performance.

Responsive Web Design: While not exclusive to HTML5, responsive web design principles are commonly applied with HTML5 and CSS3 to create Websites that adapt to different screen sizes and devices, enhancing the user experience on desktops, tablets, and smartphones.

HTML5 has become the standard for web development, providing developers with powerful tools and features to create modern, interactive, and multimedia-rich web applications and content.

V. Differences between HTML4 and HTML5

HTML5, the latest version of the Hypertext Markup Language, introduced several key features and enhancements compared to its predecessor, HTML4. Here are the main differences between HTML4 and HTML5:

Feature	HTML4	HTML5
DOCTYPE Declaration:	HTML4 requires a specific and often lengthy DOCTYPE declaration.	HTML5 has a simplified DOCTYPE declaration: <!DOCTYPE html>.
Semantic Elements:	HTML4 lacks specific semantic elements, leading developers to	HTML5 introduces semantic elements such as <header>,

	use generic elements like <code><div></code> and <code></code> for layout and structure.	<code><footer></code> , <code><article></code> , <code><section></code> , <code><nav></code> , and <code><aside></code> . These elements improve document structure and make it easier to understand the content's meaning.
Audio and Video:	HTML4 relies on third-party plugins (like Flash) for embedding audio and video content.	HTML5 provides native support for <code><audio></code> and <code><video></code> elements, allowing seamless embedding of multimedia content without plugins. It supports various formats and provides easy controls for playback.
Canvas and SVG Graphics:	HTML4 does not have native support for rendering graphics directly in the browser.	HTML5 introduces the <code><canvas></code> element for dynamic, scriptable rendering of 2D shapes and bitmap images. Additionally, SVG (Scalable Vector Graphics) is natively supported for creating vector graphics on the web.
Forms and Input Types:	HTML4 forms lack several input types available in HTML5, such as email, url, date, number, and more.	HTML5 introduces new input types and attributes, enhancing form validation and user experience. It also includes elements like <code><datalist></code> for providing predefined options for input fields.
Offline Web Applications:	HTML4 does not provide native support for offline web applications.	HTML5 includes technologies like Service Workers and the Application Cache, enabling web applications to work offline, improve performance through caching, and provide a seamless user experience even without an internet connection.
Local Storage and Session Storage:	HTML4 relies on cookies for storing limited data on the client-side.	HTML5 provides <code>localStorage</code> and <code>sessionStorage</code> APIs, allowing web applications to store data persistently or temporarily within the user's

		browser. These provide more storage and efficiency compared to traditional cookies.
Geolocation API:	HTML4 does not have a native API for accessing the user's geographical location.	HTML5 includes the Geolocation API, allowing web applications to access the user's location, facilitating location-based services and applications.
Compatibility and Support:	Older browsers might not support some advanced features of HTML4.	HTML5 is designed to be backward compatible and is widely supported by modern browsers. However, some older browsers might not fully support all HTML5 features.

HTML5's introduction of these features and enhancements has significantly improved web development, allowing developers to create more interactive, responsive, and feature-rich Websites and applications.

VI. Browser support for HTML5

HTML5, being a standard web technology, is supported by most modern web browsers. However, the level of support for specific HTML5 features can vary among different browsers. Here's an overview of the current browser support for HTML5 features. Please note that browser support might have evolved over time, so it's advisable to check the latest resources or compatibility tables for the most up-to-date information.

1. Chrome:

Google Chrome has robust support for HTML5 features. It consistently updates to support the latest HTML5 specifications.

2. Firefox:

Mozilla Firefox has strong HTML5 support and regularly updates to adhere to the latest standards. Firefox supports most HTML5 features effectively.

3. Safari:

Safari, the default browser on macOS and iOS devices, offers good support for HTML5 features. Apple keeps Safari up to date with the latest web technologies.

4. Microsoft Edge:

Microsoft Edge, the default browser on Windows 10 and later, supports HTML5 features well. It keeps up with evolving web standards.

5. Internet Explorer:

HTML5 support in older versions of Internet Explorer (such as IE 8 and below) is limited. More recent versions (IE 9 and above) have improved support but may still lack some advanced HTML5 capabilities. Microsoft has been encouraging users to switch to Microsoft Edge, which offers better HTML5 support.

6. Opera:

Opera is known for its early adoption of new web technologies. It provides good support for HTML5 features and is compatible with most web standards.

7. Mobile Browsers:

Mobile browsers on Android and iOS devices (including Chrome for Android and Safari for iOS) have excellent HTML5 support. Mobile browsers often include additional features for handling touch events and device orientation, enhancing the mobile web experience.

It's essential for web developers to be aware of specific HTML5 features they plan to use and check compatibility tables or resources like MDN Web Docs to ensure cross-browser compatibility. Additionally, as web technologies continue to evolve, it's advisable to stay up-to-date with the latest browser versions and their support for HTML5 features.

VII. Setting up your development environment

Setting up a development environment for working with HTML5 is relatively easy and straightforward since HTML5 development primarily involves a text editor and a web browser. Here are the steps to set up your development environment for HTML5:

1. Choose a Text Editor or Integrated Development Environment (IDE):

Text Editors: Visual Studio Code, Sublime Text, Atom, Notepad++ (Windows), TextMate (macOS)

Also we can use IDEs: IntelliJ IDEA, Eclipse, WebStorm

We will explain the HTML development using **Visual Studio Code**.

2. Install a Web Browser:

Chrome, Firefox, Safari, Edge, or any modern browser for testing and debugging your HTML5 applications.

3. Set Up a Local Development Server (Optional but recommended):

For simple projects, you can use the **Live Server extension** in Visual Studio Code, or similar tools available in other text editors.

For more complex projects, consider using tools like Apache, Nginx, or lite-server to serve your HTML files locally. Alternatively, you can use Node.js with Express.js for a more dynamic server setup.

4. Version Control (Optional):

Set up Git and a repository hosting service like GitHub, GitLab, or Bitbucket for version control. This is useful for tracking changes and collaborating with others.

5. Browser Developer Tools:

Familiarize yourself with the developer tools available in your chosen web browser. These tools are essential for debugging, inspecting elements, and monitoring network activity.

6. HTML5 Boilerplate (Optional but recommended for quick starting):

Consider using HTML5 Boilerplate (<https://html5boilerplate.com/>), a popular front-end template that provides a robust starting point for HTML5-based projects. It includes best practices and optimized defaults.

Depending on your project requirements, you might want to explore front-end frameworks like Bootstrap, Foundation, or CSS frameworks like Sass or LESS. JavaScript libraries like jQuery can also be useful for DOM manipulation and AJAX. Remember, the specific tools and setup can vary based on your project requirements and personal preferences. Always refer to the official documentation of the tools and frameworks you're using for detailed installation and configuration instructions.

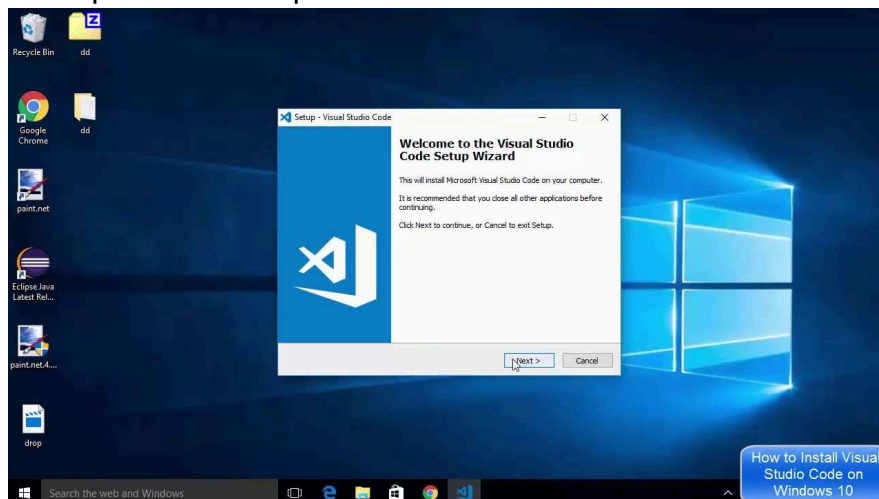
Here we will explain setting up an HTML development environment using Visual Studio Code.

Step 1 : Install Visual Studio Code on Windows.

Please visit this link and follow the instructions.

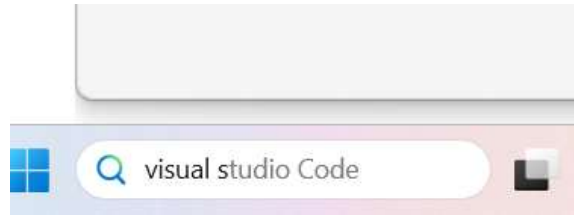
<https://code.visualstudio.com/download>

After downloading click on the installer. You will get a page like this. Just follow the instructions to complete the set up.

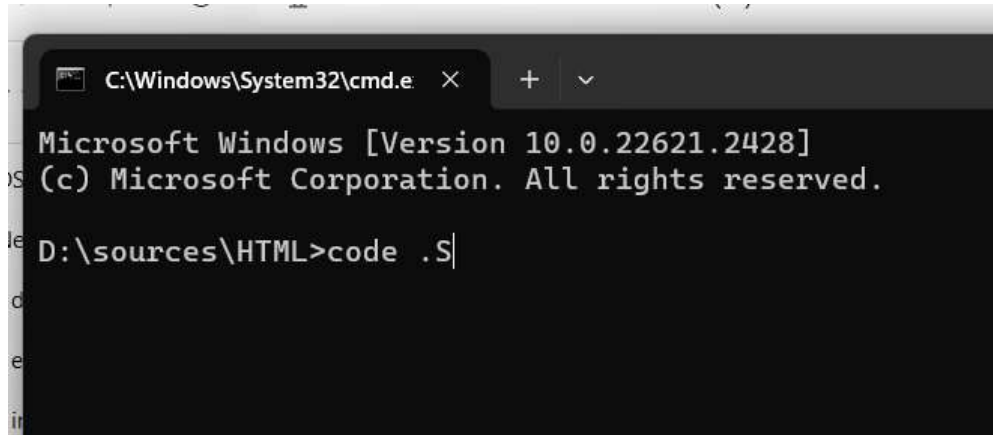


Step 2. Open the Visual Studio Editor:

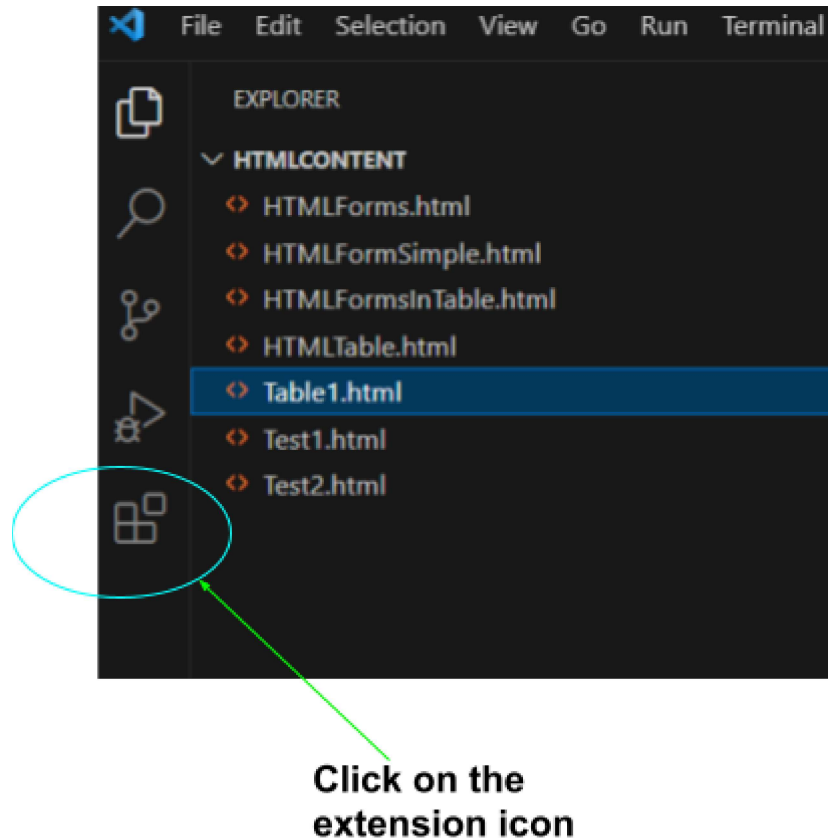
Either search from the windows search option or run the command “code .” from the source directory.



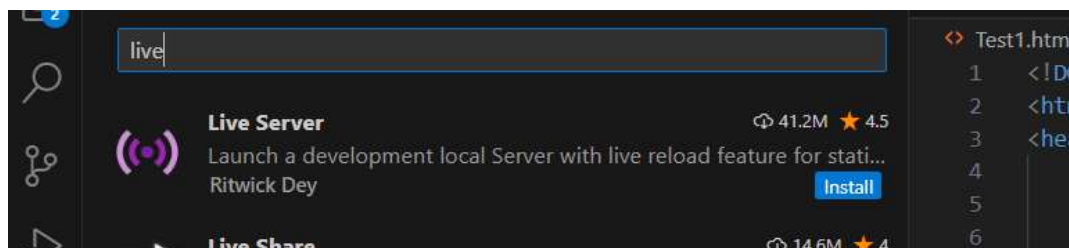
To run it from command prompt use the following command:

A screenshot of a Windows command prompt window. The title bar shows 'C:\Windows\System32\cmd.e' with a close button and a dropdown arrow. The command prompt text is as follows:
Microsoft Windows [Version 10.0.22621.2428]
(c) Microsoft Corporation. All rights reserved.
D:\sources\HTML>code .S|

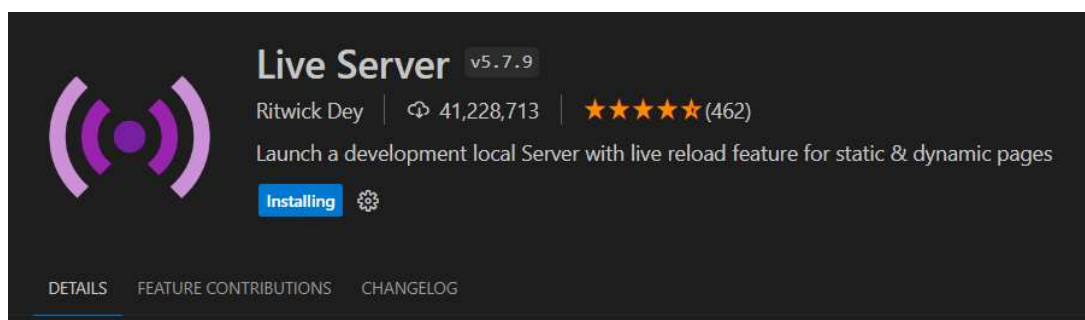
Step 3. Install Live Server extension on Visual Studio Editor:



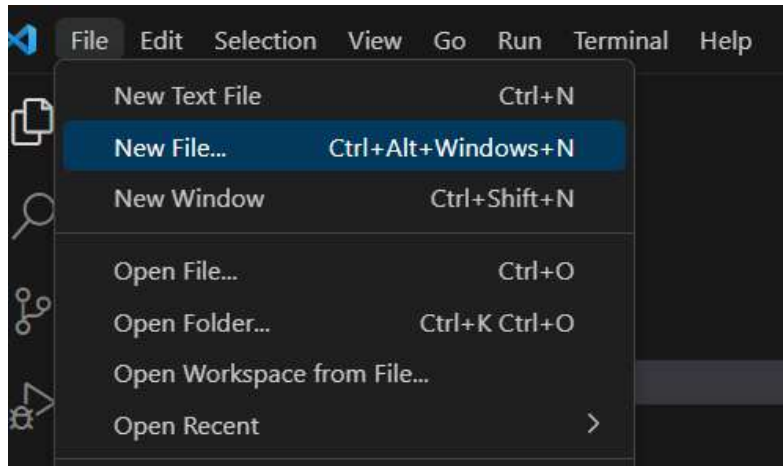
Step 4. Search for “Live Server”. Select the item and install.



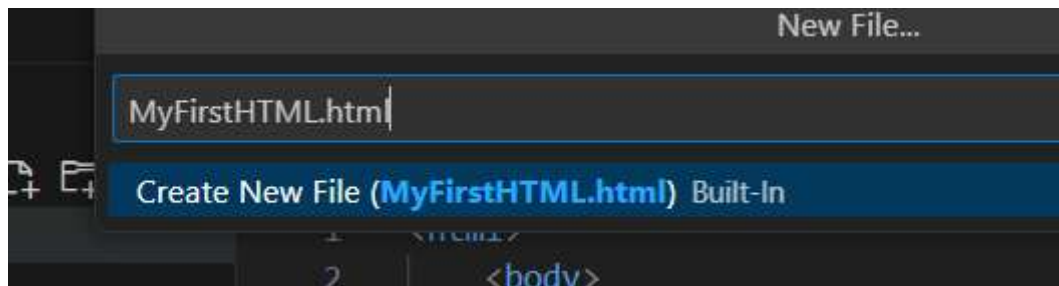
You will see the following screen while installing.



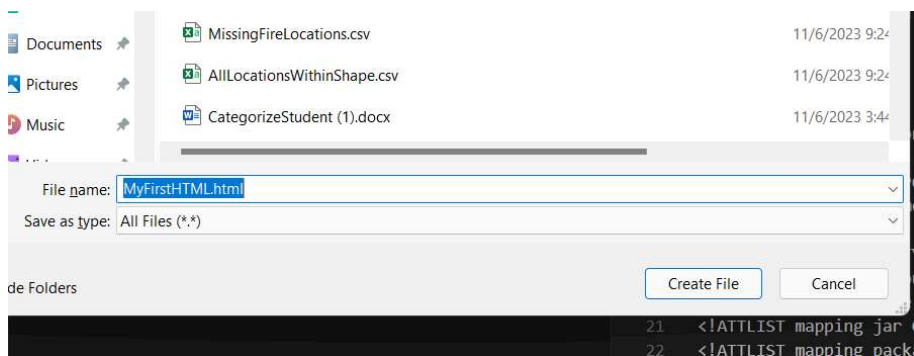
Step 5. Create a new HTML page using menu:



Give a new filename with extension.

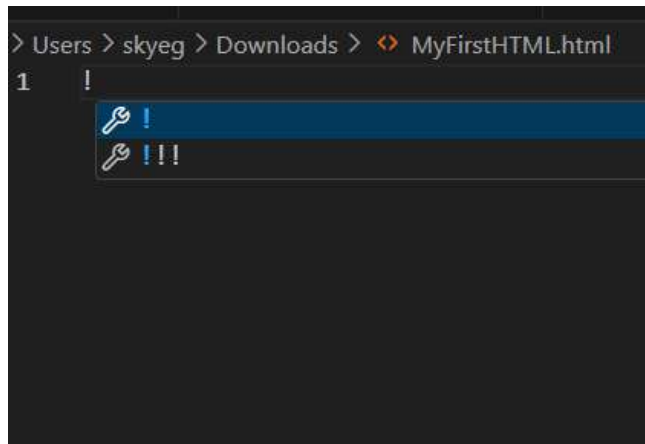


Click on the “Create File” button to create a new file in the selected folder.



Step 6. Create the HTML template code by pressing ! and Tab key.

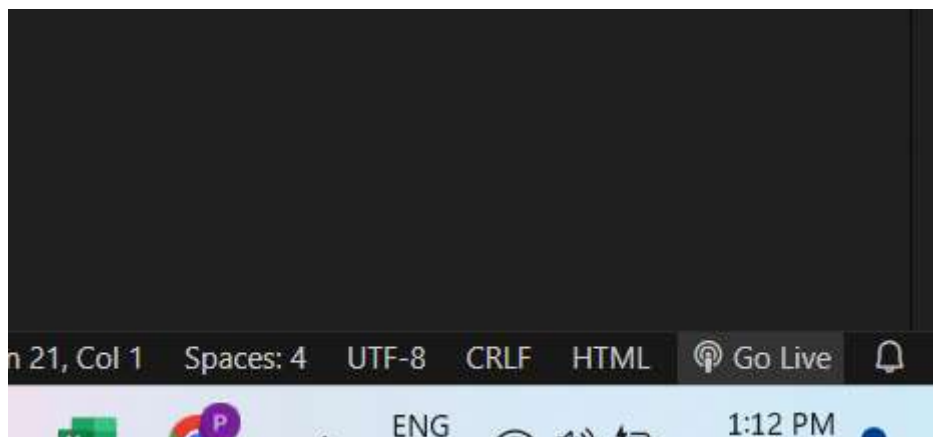
Press ! and tab key on the first line of the file.



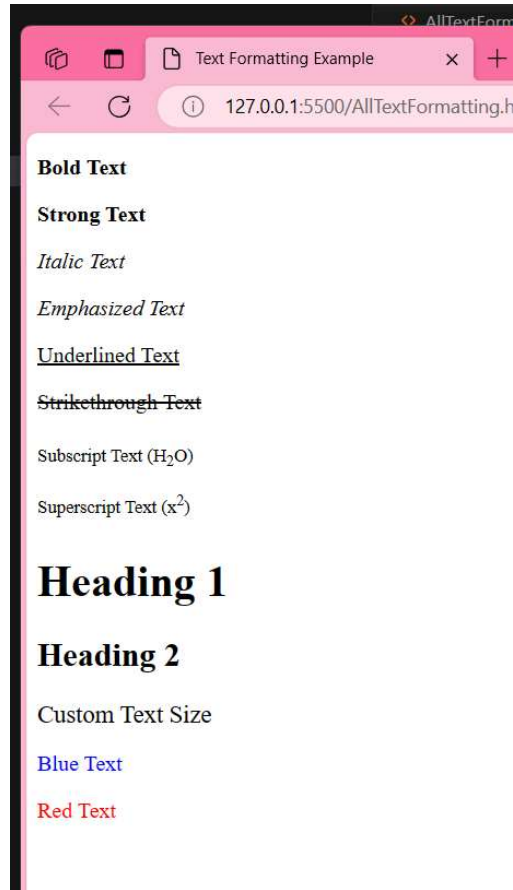
It will create basic HTML structure.

```
<!DOCTYPE html>
<html lang="en">
<head>
  <meta charset="UTF-8">
  <meta name="viewport" content="width=device-width, initial-scale=1.0">
  <title>Document</title>
</head>
<body>
  |
</body>
</html>
```

Now add some basic HTML UI items in HTML file and click on “Go Live” button.



You will see your HTML Page in the browser window.



VIII. Document Structure and Semantic Elements

Document structure in HTML refers to the overall layout and organization of an HTML5:

In HTML5, the document type declaration (DOCTYPE) is simplified and standardized to a single, concise declaration:

```
<!DOCTYPE html>
```

This declaration tells the web browser that the document is written in HTML5. Unlike previous versions of HTML, where the DOCTYPE declaration was more complex and version-specific, HTML5 uses this simplified declaration for all HTML documents.

The `<!DOCTYPE html>` declaration is placed at the very beginning of an HTML document, before the `<html>` element. It ensures that the web browser renders the document in standards mode, following the rules and specifications of HTML5.

By using this simplified DOCTYPE declaration, web developers no longer need to specify the exact version of HTML they are using. HTML5 is the latest and widely supported version of HTML, making it the default choice for modern web development.

Here are new section New Semantic Elements:

HTML5 introduces semantic elements such as `<article>`, `<section>`, `<nav>`, `<header>`, `<footer>`, `<figure>`, `<figcaption>`, `<aside>`, and `<main>`. These elements provide more meaningful and structured ways to describe the content and layout of a web page.

`<header>`, `<footer>`, `<nav>`, `<article>`, `<section>`, `<aside>`, `<main>`:

These elements help in structuring a webpage more semantically. For example, the `<header>` element typically contains introductory content or navigation links, while the `<footer>` contains the footer of a page.

`<figure>` and `<figcaption>`:

These elements are used to group and caption media content, such as images and videos. The `<figure>` element provides a container for the media, and `<figcaption>` allows you to add a caption.

`<main>`:

The **`<main>`** element represents the main content of a document. It helps screen readers and other assistive technologies identify and navigate to the primary content area.

Improved Forms:

HTML5 introduces new form elements like `<input type="date">`, `<input type="email">`, `<input type="url">`, `<input type="number">`, `<input type="range">`, and more, making it easier to capture specific types of user input.

New Attributes:

New attributes like placeholder, autocomplete, required, and others provide additional control and functionality for form elements.

`<nav>`:

The `<nav>` element is used to define a set of navigation links. It's meant to contain the primary navigation menu of a webpage.

`<section>`:

The `<section>` element is used to group related content together. It helps in organizing the content into meaningful sections.

`<article>`:

The `<article>` element represents a self-contained piece of content that could be distributed and reused independently.

<aside>:

The **<aside>** element is used for content that is tangentially related to the content around it. It is often used for sidebars or pull quotes.

Audio and Video Elements:

HTML5 introduces **<audio>** and **<video>** elements, providing native support for embedding audio and video content without the need for plugins like Flash.

<mark>:

The **<mark>** element is used to highlight or mark portions of text for reference or user attention.

These features contribute to a more structured, semantically meaningful, and accessible document structure in HTML5, allowing developers to create cleaner and more maintainable code. The improved semantic elements also aid search engine optimization (SEO) by providing clearer indications of the content's structure and meaning.

IX. HTML 5 Forms and Input Types

We already gave an introduction to HTML forms by explaining various types of form input elements in our last session. Now let us go through the additional input types provided by HTML5.

HTML5 introduced several new form elements and attributes that enhance the user experience and provide better input validation. Here are some of the new HTML5 form elements:

1. New Input Types:

<input type="email">

Allows users to enter an email address. The browser automatically validates that the input is in the form of an email address.

<input type="url">

Accepts a URL input and validates it as a valid web address.

<input type="tel">

Used for telephone numbers, providing a numeric keyboard on mobile devices for easier input.

<input type="number">

Accepts numeric input. You can set min, max, and step attributes for controlling the allowed range and increments.

<input type="date">

Provides a date picker for selecting dates.

<input type="time">

Offers a time picker for selecting times.

<input type="datetime-local">

Allows users to pick both a date and a time in the local time zone.

<input type="month">

Enables users to pick a month and year.

<input type="week">

Lets users select a specific week and year.

<input type="color">

Displays a color picker for selecting colors.

2. New Attributes:

required

Indicates that an input field must be filled out before submitting the form.

placeholder

Displays a short hint (a sample value or a short description) in the input field before the user enters a value.

pattern

Defines a regular expression pattern that the input's value must match for the form to be submitted successfully.

min and max

Specifies the minimum and maximum values for <input type="number">, <input type="range">, and date/time inputs.

step

Determines the legal number intervals for <input type="number"> and <input type="range">.

autocomplete

Controls whether the browser should autocomplete the input value based on prior user input. Values can be "on" or "off".

autofocus

Automatically focuses the input field when the page loads.

list and datalist

The list attribute associates an <input> element with a <datalist> element, providing a predefined list of options for the input field.

These new form elements and attributes enhance the functionality of HTML forms, making it easier for users to input data and for developers to validate and process that data on the client side.

X. Form basics with examples

Now let us create a HTML5 page with various HTML5 inputs and attributes:

```
<!DOCTYPE html>
<html lang="en">
<head>
  <meta charset="UTF-8">
  <meta name="viewport" content="width=device-width, initial-scale=1.0">
  <title>HTML5 Form Inputs Example</title>
  <style>
    body {
      font-family: Arial, sans-serif;
      background-color: #f4f4f4;
      margin: 0;
      padding: 0;
    }

    header {
      background-color: #2175A1;
      color: #f9f4ff;
      text-align: center;
      padding: 10px;
    }

    form {
      max-width: 600px;
      margin: 20px auto;
      padding: 20px;
      background-color: #fff;
      border-radius: 8px;
      box-shadow: 0 0 10px rgba(0, 0, 0, 0.1);
    }

    label {
      display: block;
      margin-bottom: 8px;
    }
```

```

        color: #333;
    }

    input {
        width: 100%;
        padding: 10px;
        margin-bottom: 16px;
        box-sizing: border-box;
        border: 1px solid #ccc;
        border-radius: 4px;
    }

    button {
        background-color: #0165E1;
        color: #fff;
        padding: 12px 20px;
        border: none;
        border-radius: 4px;
        cursor: pointer;
        font-size: 16px;
        transition: background-color 0.3s;
    }

    button:hover {
        background-color: #014186;
    }
</style>
</head>
<body>

<header>
    <h1>HTML5 Form Inputs Example</h1>
</header>

<form action="/submit" method="post">
    <!-- Email Input with Required and Placeholder Attributes -->
    <label for="email">Email:</label>
    <input type="email" id="email" name="email" required placeholder="Enter
your email"><br>

    <!-- URL Input with Required and Placeholder Attributes -->
    <label for="url">Website URL:</label>
    <input type="url" id="url" name="url" required placeholder="Enter Website
URL"><br>

    <!-- Telephone Input with Required and Placeholder Attributes -->

```

```
<label for="phone">Phone Number:</label>
<input type="tel" id="phone" name="phone" required placeholder="Enter
phone number"><br>

<!-- Date Input with Required, Min, and Max Attributes -->
<label for="birthdate">Birthdate:</label>
<input type="date" id="birthdate" name="birthdate" required
min="1999-01-01" max="2020-12-31"><br>

<!-- Submit Button -->
<button type="submit">Submit</button>
</form>

</body>
</html>
```

If we open the above HTML page with the browser we can see.

The image shows a web browser window with a pink header bar. The title bar indicates the page is titled "HTML5 Form Inputs Ex" and the address bar shows the URL "127.0.0.1:5500/...". The main content area has a dark blue header with the text "HTML5 Form Inputs Example". Below this, the form is displayed on a white background. It contains four labeled input fields: "Email:" with a placeholder "Enter your email", "Website URL:" with a placeholder "Enter Website URL", "Phone Number:" with a placeholder "Enter phone number", and "Birthdate:" with a placeholder "mm / dd / yyyy" and a calendar icon. A dark blue "Submit" button is located below the birthdate field. The browser's sidebar on the right shows various application icons.

In this form, we will not be allowed to enter dates out of range using the date picker. However, if the user enters a date out of range manually by writing on the input area then an error message will appear while submitting the form.

HTML5 Form Inputs Example

Email:

abc@example.com

Website URL:

http://www.example.com

Phone Number:

90808080808

Birthdate:

12/12/2023

Submit

Value must be 12/31/2020 or earlier.

To create a html form let us take a scenario of the registration page example of Websites like Amazon & Facebook. The registration page of the two Websites looks like as below:



Create account

Your name

Mobile number or email

Password

 Passwords must be at least 6 characters.

Re-enter password

Continue

By creating an account, you agree to Amazon's
[Conditions of Use](#) and [Privacy Notice](#).

Already have an account? [Sign in](#) ▶

Buying for work? [Create a free business account](#) ▶





Create a new account

It's quick and easy.

Date of birth ?

Gender ?

☐ Female☐ Male☐ Custom

People who use our service may have uploaded your contact information to Facebook. [Learn more.](#)

By clicking Sign Up, you agree to our [Terms](#), [Privacy Policy](#) and [Cookies Policy](#).
You may receive SMS notifications from us and can opt out at any time.

Sign Up

[Already have an account?](#)

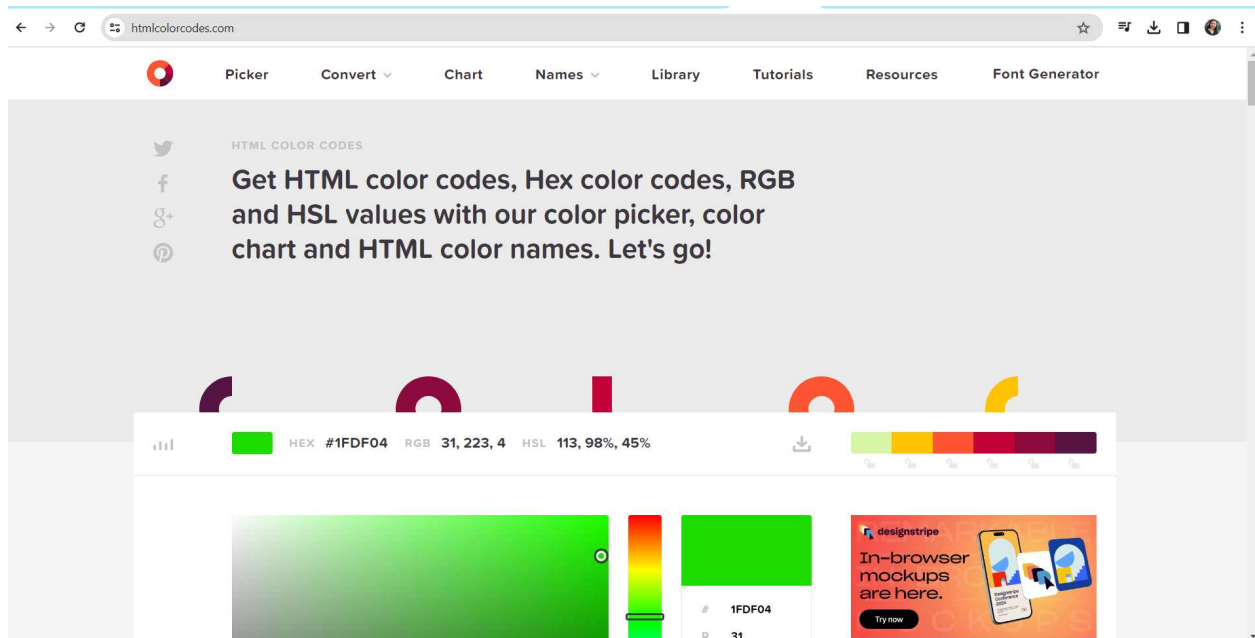
The above two registration pages are created using HTML, CSS and HTML validations.

So, let us create a similar html form for facebook registration using HTML & internal CSS.

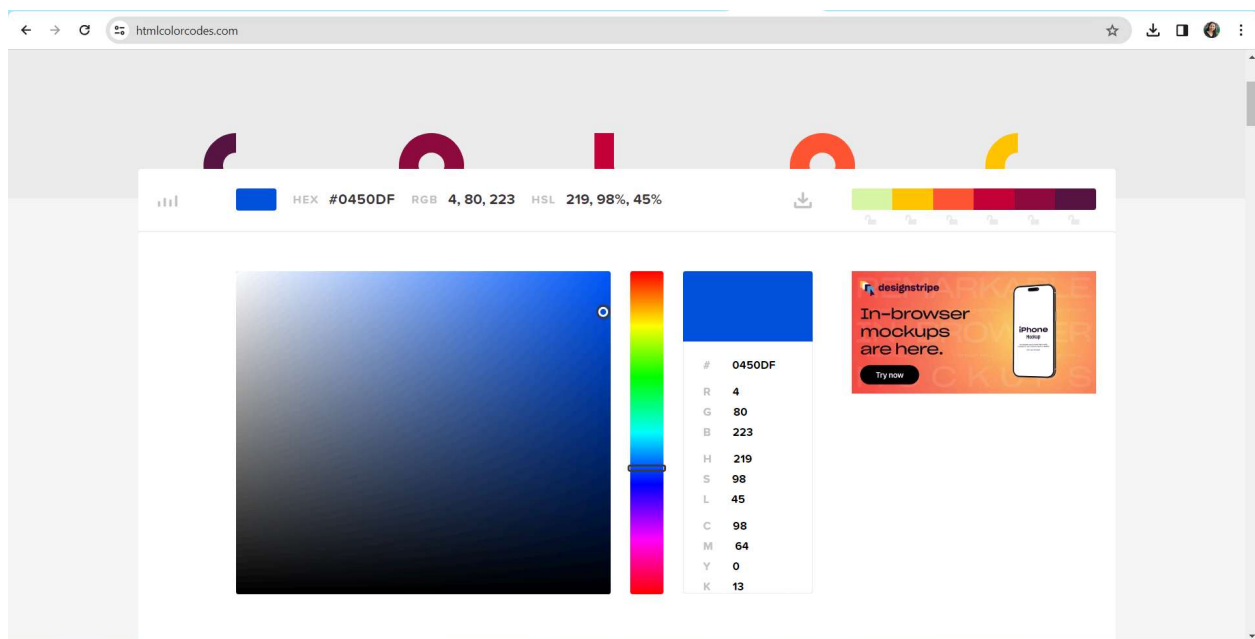
In the CSS part we will select a color code similar to the facebook registration page for the background, font color and button color.

For this we need to select a color code from the browser as follows:

1. Type html color code in the browser and open <https://htmlcolorcodes.com>. You can refer any color code scheme from the Internet.



2. From here we will select a font color for facebook heading. Drag the vertical bar from the right hand side of the page and select the dark blue color.



3. Now right click on the center square and select a similar color matching with the official facebook heading.

Copy the HEX color code that we need to use to set the color in our coding.

```
<html lang="en">
```

<head>

```
<meta charset="UTF-8">
```

```
<meta name="viewport" content="width=device-width, initial-scale=1.0">
```

```
<link rel="stylesheet"
```

```
href="https://stackpath.bootstrapcdn.com/bootstrap/4.5.2/css/bootstrap.min.css">
```

<title>HTML5 Form Inputs Example</title>

<style>

```
body {
```

font-family: Arial, sans-serif;

```
background-color: #f4f4f4;
```

```
margin: 0;
```

padding: 0;

}

```
header {
```

```
background-color: #2175A1;
```

```
color: #f9f4ff;
```

```
text-align: center;
```

padding: 10px;

}

form {

```
max-width: 600px;
```

```
margin: 20px auto;
```

padding: 20px;

background-color: #fff;

border-radius: 8px;

```
box-shadow: 0 0 10px rgba(0, 0, 0, 0.1);
```

}

label {

```
display: block;
```

```
margin-bottom: 8px;
```

```
color: #333;
```

}

```
input {
```

```
width: 100%;
```

padding: 10px;

```
margin-bottom: 16px;
```

box-sizing: border-box;

border: 1px solid #ccc;

border-radius: 4px;

```

    }

    /* Button styles */
.submit-btn {
padding: 5px 50px;
border: none;
border-radius: 4px;
background-color: #008000;
color: #fff;
cursor: pointer;
transition: background-color 0.3s ease-in-out;
margin-left: 30%;
}

.submit-btn:hover {
background-color: #008000;
}

/* Additional styling for radio buttons */
.custom-control-input:checked~.custom-control-label::before {
background-color: #4CAF50;
border-color: #4CAF50;
}

.custom-control-input:focus~.custom-control-label::before {
box-shadow: 0 0 0 1px #fff, 0 0 0 3px #45a049;
}

.title{
color:#316FF6;
text-align: center;
}

.top
{

text-align: center;
font-size: 14px;
}

p
{
font-size: 11px;
}

.container

```

```

{
    width: 30%;
}
</style>
</head>
<body>
    <div class="container">
        <form action="/submit" method="post">
            <div class="title">
                <h1><b>facebook</b></h1>
            </div>
            <div class="row-md-2">
                <div class="top">
                    <h4><b>Create Account</b></h4>
                    <span>It's quick and easy.</span>
                </div>
            </div>
            <hr>

            <div class="row">
                <div class="col-md-6">
                    <!-- Name validation -->
                    <input type="text" class="form-control" placeholder="First name" required>
                </div>
                <div class="col-md-6">
                    <input type="text" class="form-control" placeholder="Surname" required>
                </div>
            </div>

            <div class="row-md-2">
                <!-- Email input validation -->
                <input type="text" class="form-control" placeholder="Mobile number or email address" required>
            </div>

            <div class="row-md-2">
                <!-- password validation -->
                <input type="password" class="form-control" placeholder="New password" required minlength="6" maxlength="20">
            </div>

            <div class="row-md-2">
                <span>Date Of Birth</span>
                <div class="form-row">
                    <!-- Day Dropdown -->
                    <div class="form-group col-md-3">

```

```
<select id="day" name="day" class="form-control">
  <option value="" disabled selected>Day</option>
  <option value="1">1</option>
  <option value="2">2</option>
  <option value="3">3</option>
  <option value="4">4</option>
  <option value="5">5</option>
  <option value="6">6</option>
  <option value="7">7</option>
  <option value="8">8</option>
  <option value="9">9</option>
  <option value="10">10</option>
</select>
</div>

<!-- Month Dropdown -->
<div class="form-group col-md-5">
  <select id="month" name="month" class="form-control">
    <option value="" disabled selected>Month</option>
    <option value="1">Jan</option>
    <option value="2">Feb</option>
    <option value="3">Mar</option>
    <option value="4">Apr</option>
    <option value="5">May</option>
    <option value="6">Jun</option>
    <option value="7">Jul</option>
    <option value="8">Aug</option>
    <option value="9">Sep</option>
    <option value="10">Oct</option>
    <option value="11">Nov</option>
    <option value="12" selected="1">Dec</option>
  </select>
</div>

<!-- Year Dropdown -->
<div class="form-group col-md-4">
  <select id="year" name="year" class="form-control">
    <option value="" disabled selected>Year</option>
    <option value="2023" selected="1">2023</option>
    <option value="2022">2022</option>
    <option value="2021">2021</option>
    <option value="2020">2020</option>
    <option value="2019">2019</option>
    <option value="2018">2018</option>
    <option value="2017">2017</option>
    <option value="2016">2016</option>
```

```

        <option value="2015">2015</option>
        <option value="2014">2014</option>
        <option value="2013">2013</option>
        <option value="2012">2012</option>
        <option value="2011">2011</option>
        <option value="2010">2010</option>
    </select>
</div>
</div>
</div>

<div class=" row-md-2">
    <span>Gender</span>
    <div class="form-row">
        <div class="form-group col-md-12">
            <div class="custom-control custom-radio custom-control-inline">
                <input type="radio" id="male" name="gender"
class="custom-control-input" value="male">
                <label class="custom-control-label" for="male">Male</label>
            </div>
            <div class="custom-control custom-radio custom-control-inline">
                <input type="radio" id="female" name="gender"
class="custom-control-input" value="female">
                <label class="custom-control-label" for="female">Female</label>
            </div>
            <div class="custom-control custom-radio custom-control-inline">
                <input type="radio" id="other" name="gender"
class="custom-control-input" value="other">
                <label class="custom-control-label" for="other">Other</label>
            </div>
        </div>
    </div>
</div>

<div class=" row-md-2">
    <p>People who use our service may have uploaded your contact information
to Facebook.
        <a href="https://www.facebook.com/help/637205020878504">Learn
more.</a></p>
    </div>

<div class=" row-md-2">
    <p>By clicking Sign Up, you agree to our
        <a href="https://www.facebook.com/legal/terms/update">Terms</a>,

```

```

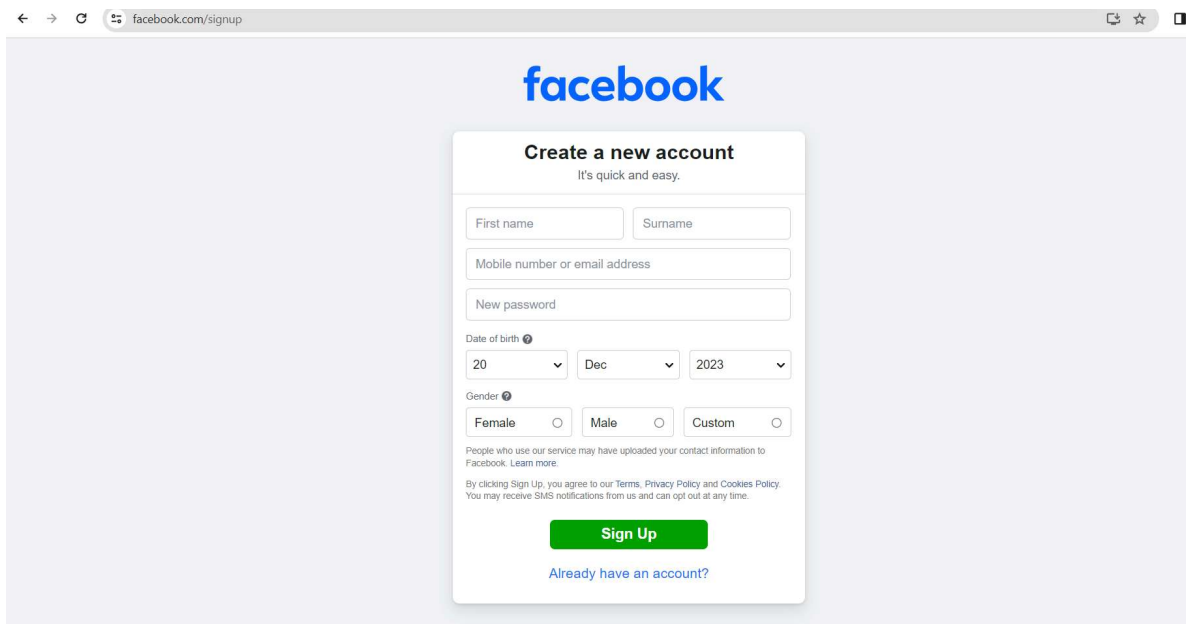
        <a
href="https://www.facebook.com/privacy/policy/?entry_point=data_policy_redirect&entry
=0">Privacy Policy</a>
        and <a
href="https://www.facebook.com/privacy/policies/cookies/?entry_point=cookie_policy_re
direct&entry=0">Cookies Policy.</a>
        You may receive SMS notifications from us and can opt out at any
time.</p>
    </div>
    <!-- Submit Button -->
    <button type="submit" class="submit-btn"><b>Sign Up</b></button>
</form>
</div>
</div>

</body>

</html>

```

Output:



The screenshot shows the Facebook sign-up page in a web browser. The URL bar displays 'facebook.com/signup'. The page features the Facebook logo at the top. Below it, a white card contains the heading 'Create a new account' with the subtext 'It's quick and easy.' The form includes several input fields: 'First name', 'Surname', 'Mobile number or email address', and 'New password'. Below these is a 'Date of birth' section with three dropdown menus set to '20', 'Dec', and '2023'. A 'Gender' section follows with three radio button options: 'Female', 'Male', and 'Custom'. At the bottom of the form, there is a green 'Sign Up' button and a link that says 'Already have an account?'. Small text at the bottom of the form mentions contact information uploads and terms of service.

In this form validations are added for all the text fields.

facebook

Create Account

It's quick and easy.

First name Surname

Please fill out this field. address

New password

Date Of Birth

Day Dec 2023

Gender

☐ Male ☐ Female ☐ Other

People who use our service may have uploaded your contact information to Facebook. [Learn more](#).

By clicking Sign Up, you agree to our [Terms](#), [Privacy Policy](#) and [Cookies Policy](#). You may receive SMS notifications from us and can opt out at any time.

Sign Up

ChatGPT Exercise:

Please visit the site - <https://chat.openai.com/> and create your account if not yet created. Next, start with the first question on the prompt:

I have an old HTML page developed using HTML4. I want to improve the page's appearance using HTML5. What should I do?

I have an old HTML page developed using HTML4. I want to improve the page's appearance using HTML5. What should I do?



Submit the query clicking the green arrow icon. Now go through the response received from the AI engine. Output will be something like the following screenshot.



Transitioning an old HTML page developed using HTML4 to HTML5 involves updating the markup to use HTML5 features and semantic elements. Here's a step-by-step guide to improving your page's appearance and structure using HTML5:



1. Update the Document Type Declaration:

Replace the old DOCTYPE declaration with the HTML5 standard:

html

Copy code

```
<!DOCTYPE html>
```

2. Replace Deprecated Elements:

Replace deprecated elements from HTML4 with modern alternatives:

- Replace `**<font>**` tags with CSS for text styling.
- Replace `**<center>**` tags with CSS for centering elements.
- Replace `**<strike>**` tags with `**<del>**` or CSS for strikethrough effects.
- Replace `**<b>**` and `**<i>**` tags with `**<strong>**` and `**<em>**` for semantic importance.

3. Use Semantic Elements:

Regenerate

After the 2 hr ILT, the student has to do a 1 hour live lab. Please refer to the Session 4 Lab doc in the LMS.